

Quantum Information and Computation

Structure-only skeleton (course content pending)

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2026-09-23

Table of contents

1	About these notes	1
1.1	Scope	1
1.2	Sources (planned)	1
2	Qubits and single-qubit operations	3
2.1	Planned outline	3
2.2	References	3
3	Multi Qubit Systems	5
3.1	Planned outline	5
3.2	References	5
4	Quantum Circuits	7
4.1	Planned outline	7
4.2	References	7
5	Quantum Algorithms	9
5.1	Planned outline	9
5.2	References	9
6	Quantum Error Correction	11
6.1	Planned outline	11
6.2	References	11
7	Quantum Information Theory	13
7.1	Planned outline	13
7.2	References	13

1 About these notes

1.1 Scope

These are lecture notes for *Quantum Information and Computation* (third-semester elective, Master in Physics of Data, University of Padova). At the time of writing the professor has not yet distributed lecture slides; this book is therefore a **structural skeleton** — chapter titles and a suggested organisation that will be filled in as the course progresses.

The skeleton follows the standard Nielsen–Chuang curriculum:

- Chapter 1 — single-qubit systems, Bloch sphere, and single-qubit gates.
- Chapter 2 — multi-qubit systems, entanglement, and Bell states.
- Chapter 3 — quantum circuits and universal gate sets.
- Chapter 4 — quantum algorithms (Deutsch–Jozsa, Grover, Shor).
- Chapter 5 — quantum error correction.
- Chapter 6 — quantum information theory (von Neumann entropy, quantum channels).

Every chapter is currently a stub with a suggested table of contents; it will be expanded once the course delivers content.

1.2 Sources (planned)

Primary reference: Nielsen & Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge (2010). Secondary: Benenti, Casati & Strini, *Principles of Quantum Computation and Information*, World Scientific (2004).

2 Qubits and single-qubit operations

Status: skeleton. This chapter is a placeholder; content will be added once the course delivers lecture material.

2.1 Planned outline

- Postulates of quantum mechanics for a two-level system.
- The Bloch sphere and geometric visualisation.
- Pauli operators, Hadamard, phase, and rotation gates.
- Measurement in a general basis; projective vs POVM.
- Physical implementations: superconducting transmon, ion trap, neutral atom (link to 2sem IQH notes).

2.2 References

Nielsen & Chuang, Chapters 1–2. Benenti, Casati & Strini, Chapters 2–3.

3 Multi Qubit Systems

Status: skeleton. Content will be added once the course delivers lecture material.

3.1 Planned outline

(to be filled in)

3.2 References

Nielsen & Chuang; Benenti, Casati & Strini.

4 Quantum Circuits

Status: skeleton. Content will be added once the course delivers lecture material.

4.1 Planned outline

(to be filled in)

4.2 References

Nielsen & Chuang; Benenti, Casati & Strini.

5 Quantum Algorithms

Status: skeleton. Content will be added once the course delivers lecture material.

5.1 Planned outline

(to be filled in)

5.2 References

Nielsen & Chuang; Benenti, Casati & Strini.

6 Quantum Error Correction

Status: skeleton. Content will be added once the course delivers lecture material.

6.1 Planned outline

(to be filled in)

6.2 References

Nielsen & Chuang; Benenti, Casati & Strini.

7 Quantum Information Theory

Status: skeleton. Content will be added once the course delivers lecture material.

7.1 Planned outline

(to be filled in)

7.2 References

Nielsen & Chuang; Benenti, Casati & Strini.

